Keivalya Pandya

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Education

Northeastern University **Dec 2026**
Master of Science, Robotics (Computer Science) (GPA: 3.952)
• **Coursework:** Deep Learning, Mechanistic Interpretability, Reinforcement Learning, Master's Research (Imitation Learning)

Birla Vishvakarma Mahavidyalaya **Jun 2023**
Bachelor of Technology, Mechanical Engineering
• **Achievements:** Secured State Government Scholarship Tuition Fee Waiver

Work Experience

General Motors | *Robotics Engineering Intern* **May 2026 - Aug 2026**
• Worked on vision-language-action (VLA) and world-action-model (WAM) approaches for bimanual manufacturing manipulation; contributed to data collection, preprocessing, and ablation studies on a UR20/YAM-arm setup.
• Benchmarked 7 open-source models across simulation and hardware scenarios; evaluated behavior in challenging operating conditions. Audited real-world UR20 manipulation data (269 demonstrations, 105K frames, 3 camera views); analyzed workspace coverage and action-prediction errors to prioritize targeted data collection
• Conducted ACT action-chunking ablations, increasing task success from 0.5% to 38.5% without new data or architecture changes; analyzed grasp initiation, box drops, and stability failures.

Familiar Machines & Magic | *Reinforcement Learning Intern* **Sep 2025 - Dec 2025**
• Designed, implemented, and tested learning algorithms using IsaacLab and sim2real techniques to enhance autonomous robotic behaviors, resulting in improved performance in real-world environments
• Worked with AI, Hardware, and Software teams to develop embedded firmware, unit tests, and simulation scripts, enhancing policy optimization and improving robot efficiency

Northeastern University | *Robotics AI Researcher* **May 2025 - Aug 2025**
• Developed a human-in-the-loop reinforcement-learning framework in Isaac Sim for robot participation in ongoing human interactions; implemented spatial-heuristic pose proposals and an MLP-based pose-scoring pipeline.

Research Apprentice **Jan 2025 - Apr 2025**
• Deployed and compared Diffusion Policy, VQ-BeT, and ACT on Hello Robot Stretch 3; investigated OpenVLA/RUM deployment. Selected for a funded Khoury research apprenticeship (40 of 20,000+ students).

Projects

pico-WAM
• Built a 1.33M latent world-action model with action-conditioned dynamics, state/image prediction, and model-based planning; trained using 206 human Push-T demonstrations plus simulator-branched transitions.
• Achieved 144/300 successes on unseen Push-T episodes with planning vs. 110/300 for a behavioral-cloning baseline across 3 runs; documented action-head, data-coverage, and planning ablations.

mini-VLA
• Built an end-to-end Vision-Language-Action policy that learns instruction-conditioned robot manipulation from demonstrations using diffusion-based action generation.
• Integrated vision, language, and proprioceptive encoders into a lightweight (~150 LOC core), modular framework enabling fast iteration and research prototyping in simulation.

Determining Language Grounding Failures in Vision-Language-Action Models
• Designed scene-fixed language-intervention evaluations of VLA policies up to 7.5B parameters; observed a 31.7-percentage-point performance drop under contrastive action prompts.

RoboSpace.app
• Built a browser-based robot simulator; reported 1,000+ unique visitors and 200+ registrations within 36 hours of launch. Serving 1200+ active users.

Skills

• Python, C++, Pytorch, ROS2, Docker, VLA/WAM, Imitation Learning, Reinforcement Learning, MuJoCo, Isaac Sim/Lab, Gazebo, OpenCV, Manipulation, Controls, Real-to-Sim

Publications

• Pandya K.AI-based Bio-electro-mechanical Prosthetic Limb..*IN Patent 356336-001/125869*